

Optimising Pairs Trading through Dynamic Screening,
Adaptive Mean-Reversion Windows and Kelly Criterion
Capital Allocation

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Abstract

Pairs trading identifies related securities and trades on divergence using a market-neutral approach (Gatev, Goetzmann, & Rouwenhorst, 2006). We examine whether adapting this framework through continuous relationship validation, signal timing aligned to reversion speed, and edge-based capital allocation generates positive risk-adjusted returns. We construct a backtesting system using rolling cointegration tests, adaptive Z-score windows, and Kelly Criterion position sizing. Results demonstrate that statistically grounded adaptivity improves risk-adjusted returns in non-stationary markets.

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1. Introduction

1.1 Background

Pairs trading capitalises on mispricing in cointegrated pairs by modeling stationary spreads and trading temporary divergences under mean-reversion expectations (Gatev et al., 2006). This market-neutral strategy was developed by Morgan Stanley in the mid-1980s.

1.2 Research Objective

Traditional pairs trading assumes stable cointegration, yet relationships decay due to fundamentals and macroeconomic events. Fixed Z-score windows assume uniform reversion rates, and fixed position sizes overlook volatility differences. We address these through dynamic rescreening using rolling cointegration tests, adaptive Z-score windows calibrated to each pair's half-life, and Kelly Criterion position sizing. Using minute-level U.S. equity data with transaction costs, results show considerable gains in Sharpe and Sortino ratios, demonstrating that principled adaptivity improves performance.

2. Data Description

2.1.1 Data Sources & Universe

We obtained 1-minute OHLCV equity data from Databento. Our universe consists of 21 liquid, large-cap U.S. equity pairs from the same sectors or with established economic linkages. Our sample period spans 17/10/2020-17/10/2025, covering regular trading hours only.

2.1.2 Data Quality & Processing

The validation pipeline implements quality controls. Missing bars are forward-filled using the last known price, avoiding interpolation to prevent look-ahead bias.

2.1.3 Potential Data Limitations

Our universe consists of current index constituents, introducing survivorship bias by excluding delisted, bankrupt, or acquired firms. Transaction cost assumptions are simplified approximations that cannot capture order size relative to volume, spread width, or adverse selection. These are standard high-frequency limitations unlikely to materially impact qualitative conclusions, though absolute performance metrics require appropriate uncertainty.

3. Baseline Strategy

3.1 Theoretical Foundation

Cointegration describes a long-run equilibrium between two non-stationary price series whose linear combination is stationary. Formally:

$$Y_t = \alpha + \beta X_t + u_t$$

Where u_t is the stationary spread fluctuating around a constant mean. When the spread deviates, economic forces restore balance. We profit by shorting the overpriced leg and longing the underpriced one.

3.2 Base Methodology

Step 1: Cointegration Testing

We apply three stationarity tests. ADF tests for unit roots (rejection at $p < 0.05$ indicates stationarity). Phillips Perron (PP) provides similar results with greater robustness to heteroskedasticity

and autocorrelation. Variance Ratio (VR) confirms mean reversion by checking that variance scales sublinearly with time, expecting $VR(k) < 1$ at horizons $k = 2, 4, 8$.

Only pairs satisfying all three tests are selected. Screening occurs once at the start of each trading period, with relationships assumed valid throughout that window.

Step 2: Spread Construction

For each accepted pair:

$$u_t = Y_t - (\alpha + \beta X_t)$$

We standardise to a Z-score

$$z_t = \frac{u_t - \mu_u}{\sigma_u},$$

where μ_u and σ_u are the mean and standard deviation of the spread of a fixed rolling window (e.g, 1000 min).

Step 3: Entry/Exit Rules

- Entry: when $2 < |z_t| < 3$
- Exit: when $|z_t| < 0.5$ (mean reversion)
- Stop-loss: when $|z_t| > 4$ (adverse movement)

Step 4: Position Sizing

Fixed notional per trade (e.g. \$10,000).

Quantities are beta-hedged:

$$q_Y = \frac{\text{notional}}{P_Y}, \quad q_X = -\beta q_Y$$

3.3 Data & Assumptions

Universe: Sector-matched large-cap U.S. equities (e.g., XOM/CVX, V/MA, KO/PEP, JPM/BAC). For the full table, see Appendix, table A1.

Sample Period: Train set (01/11/2020-01/10/2023) for parameter tuning; test set (02/10/2023-01/10/2025) for out-of-sample validation.

Transaction Costs: 1 bps per trip (0.5 bps execution + 0.5 bps slippage).

Initial Capital: \$100,000.

3.4 Baseline Performance

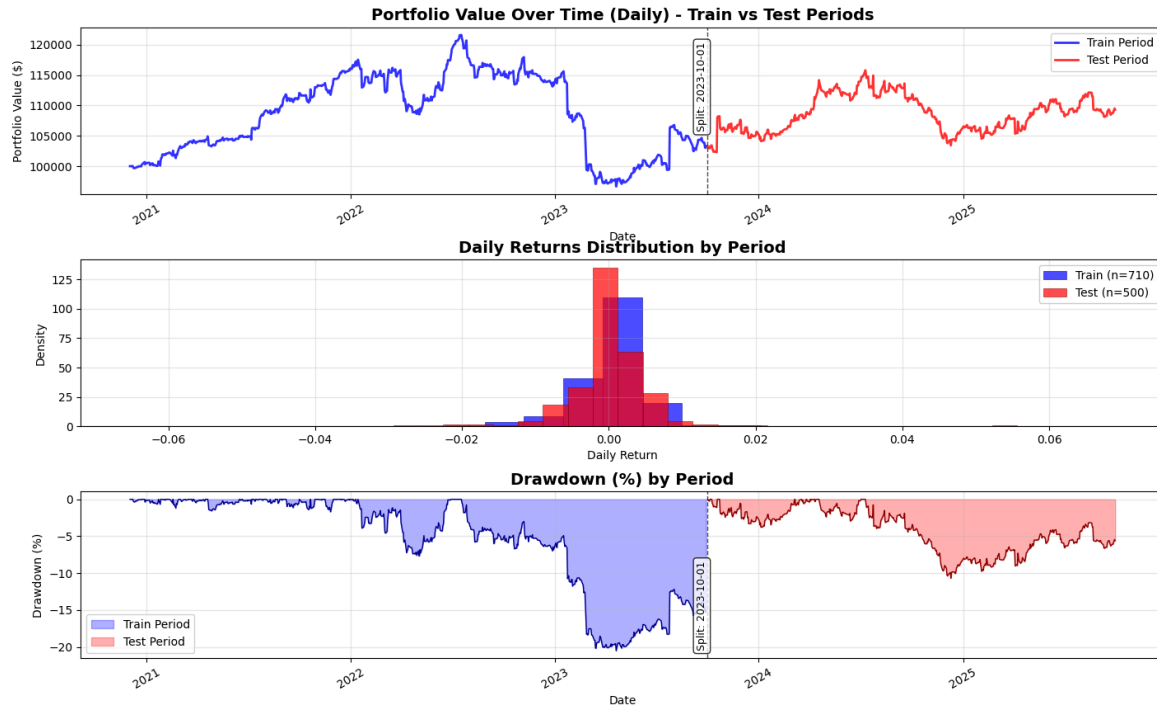


Figure 1. Portfolio value (\$), Daily returns, Drawdown (%) of Baseline Model

Metric	Train	Test	Combined
Total Return (%)	3.31	5.93	9.31
CAGR (%)	1.16	2.94	1.87
Sharpe Ratio	-0.35	-0.20	-0.29
Sortino Ratio	-0.38	-0.25	-0.34
Max Drawdown (%)	-20.53	-10.69	-20.53
Volatility (%)	9.35	8.32	8.93
Win Rate (%)	52.5	50.8	51.8

Table 1. Train, Test, and Combined Metrics of Baseline Model

The base strategy delivers small gains but no meaningful risk-adjusted performance. As observed in Table 1 and Figure 1, training achieved 3.3% return (Sharpe -0.35), while testing improved to 5.9% (Sharpe -0.20), yielding 1.9% CAGR, suggesting random returns. The 52% win rate masks unfavorable

payoffs where small profits are offset by larger losses. The equity curve experiences a 20% drawdown during 2022–2023, followed by persistent 10% drawdowns, indicating regime decay. Daily returns cluster around zero with heavier negative tails, with occasional large losses dominating frequent small wins.

4. Weaknesses and Proposed Improvements

4.1 Weakness #1: Static Cointegration Relationships

4.1.1 Problem Statement

The base approach screens for cointegration once, assuming relationships hold throughout trading. However, cointegration degrades through structural breaks, regulatory changes, and macro shocks. Trading stale relationships causes repeated losses from expired mean-reversion. Since events occur monthly or quarterly, cointegration must be re-validated regularly (e.g., every 30 days).

4.1.2 Empirical Evidence: XOM-CVX Case Study

To illustrate cointegration instability, consider Exxon Mobil (XOM) and Chevron (CVX), two integrated oil majors. During August-September 2023, the pair exhibited strong cointegration (ADF p-value 0.007, below the 0.05 threshold, see Figure 2).

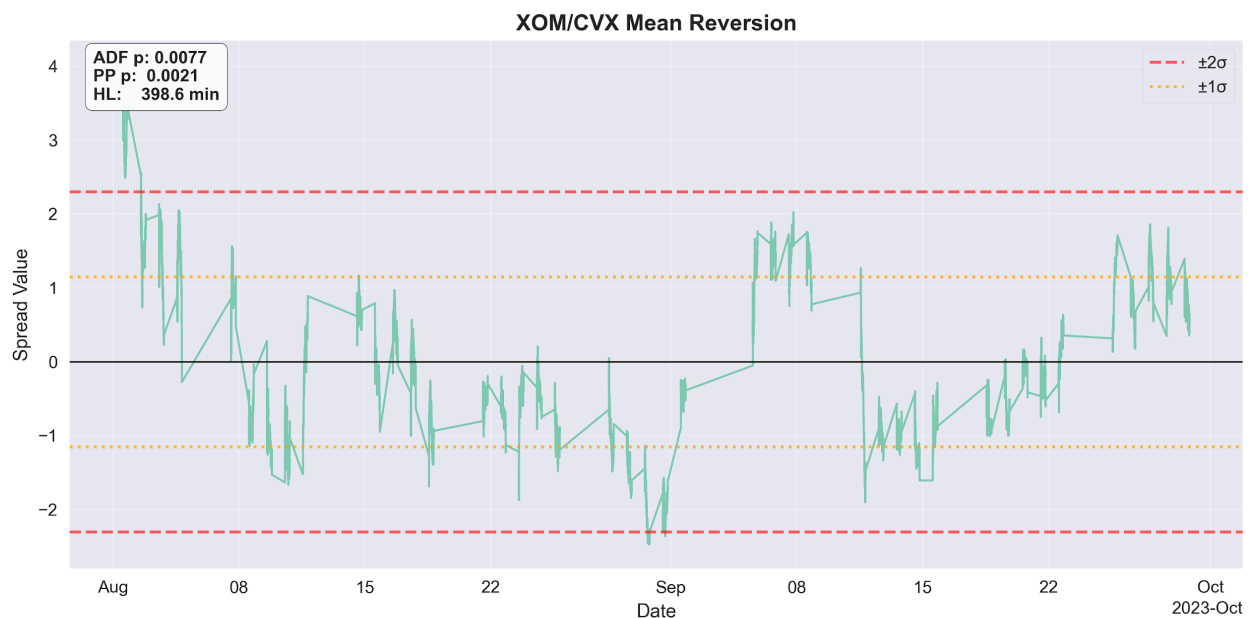


Figure 2. XOM/CVX Spread, August to October

By October 2023, the relationship collapsed when Exxon acquired Pioneer Natural Resources (\$59.5B) and Chevron acquired Hess Corporation (\$53B), asymmetrically altering risk profiles (see Figure 3). The ADF p-value rose to 0.95, and the spread trended directionally. Trading with prior parameters would have generated sustained losses, highlighting the need for frequent re-screening.

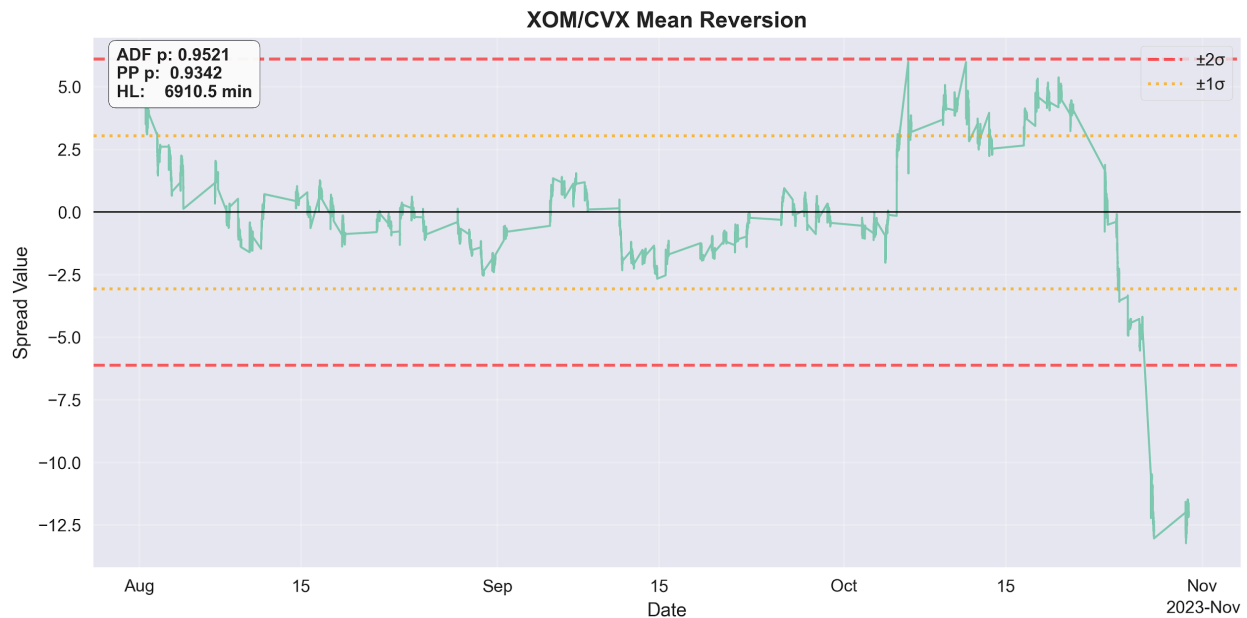


Figure 3. XOM/CVX Spread, August to November

4.1.3 Proposed Solution: Dynamic Rolling Rescreening

We replace single screening with rolling re-testing that treats cointegration as renewable.

1. Frequent Rescreening

Each epoch uses a 30-day rolling window to estimate relationships, then trades forward 30 days before re-screening.

2. Adaptive Parameters

Every epoch re-estimates parameters from the most recent 30 days, ensuring hedge ratios reflect current sensitivities

3. Descreen Logic

Pairs failing cointegration re-tests trigger immediate position exits, preventing losses from structural breaks.

4.1.4 Incremental Results

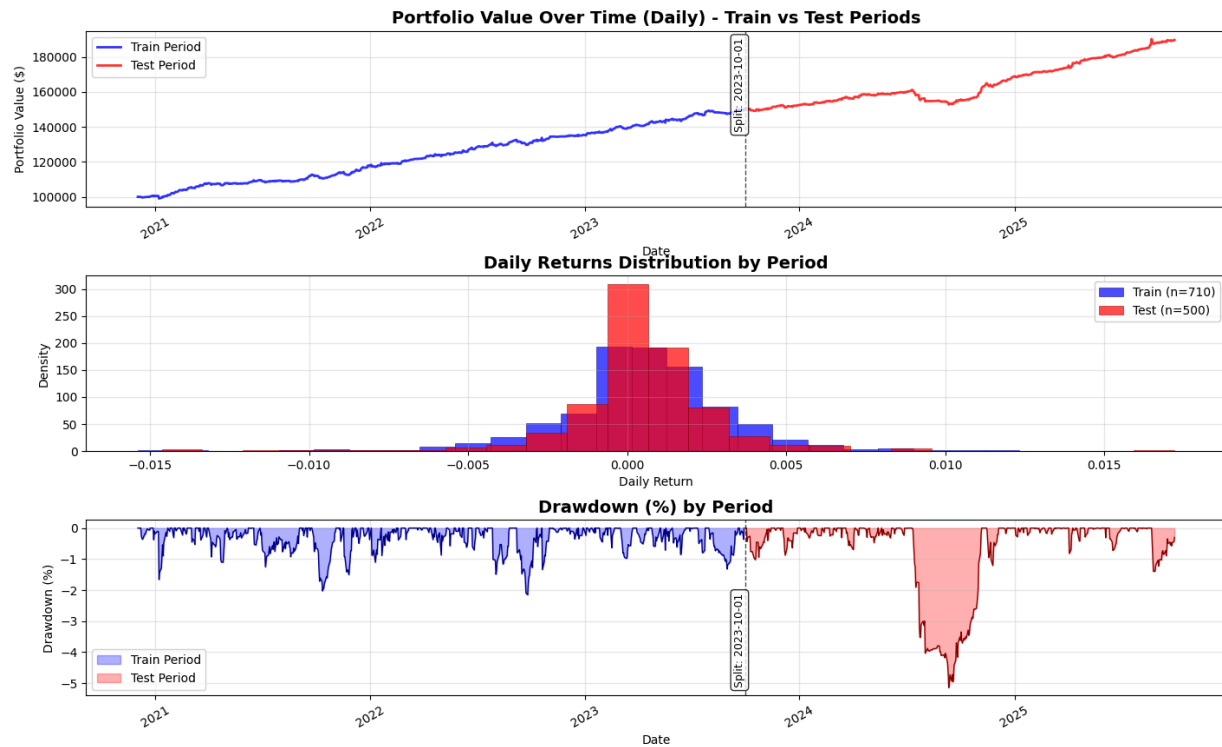


Figure 4. Portfolio value (\$), Daily returns, Drawdown (%) of Base + Re-screening Model

Base + Rescreening Performance (Train vs Test)

Metric	Train	Test	Combined
Total Return (%)	50.08	26.07	89.67
CAGR (%)	15.48	12.36	14.24
Sharpe Ratio	2.24	1.80	2.07
Sortino Ratio	2.82	2.05	2.51
Max Drawdown (%)	-2.15	-5.14	-5.14
Volatility (%)	4.31	3.82	4.11
Win Rate (%)	61.1	60.2	60.8

Table 2. Train, Test, and Combined Metrics of Base + Re-screening Model

Combined (Train + Test) Performance (Base vs Base + Rescreening)

Metric	Base	Base + Rescreening
Total Return (%)	9.31	89.67
CAGR (%)	1.87	14.24
Sharpe Ratio	-0.29	2.07
Sortino Ratio	-0.34	2.51
Max Drawdown (%)	-20.53	-5.14
Volatility (%)	8.93	4.11
Win Rate (%)	51.8	60.8

Table 3. Baseline vs Base + Re-screening Metrics

Monthly dynamic re-screening produces substantial improvements (see Figure 4). Combined return rises from 9.3% to 89.7% (CAGR 1.9% to 14.2%), with Sharpe increasing from -0.29 to 2.07 (Table 2). Win rate climbs from 51.8% to 60.8%, while max drawdown improves from 20.5% to 5.1% and volatility halves from 8.9% to 4.1% (Table 3). Frequent re-screening prevents stale relationships, ensuring only cointegrated pairs are traded and eliminating regime drift.

4.2 Weakness #2: Fixed Z-Score Lookback Window

4.2.1 Problem Statement

The base model uses a uniform 1000-minute window, assuming identical reversion speeds. Fast pairs (ETFs, high-volume tech) revert within minutes while slow pairs (industrials, utilities) revert over days. Fixed windows fail both: long windows cause late entries for fast pairs, and short windows trigger false signals for slow pairs. The solution is tailoring lookback windows to each pair's reversion speed.

4.2.2 Empirical Examples

To illustrate the discrepancies between reversion speeds of different assets, we look at real estate ETFs (IYR/VNQ) and energy stocks (XOM/CVX) over the period Jan to Mar 2025 (Figure 5).

Example 1: Fast Pair (IYR–VNQ)

The spread had a half-life of 20 minutes. A 1000-minute window remembers nearly 10 hours, smoothing short-term moves and triggering signals after reversion occurs.

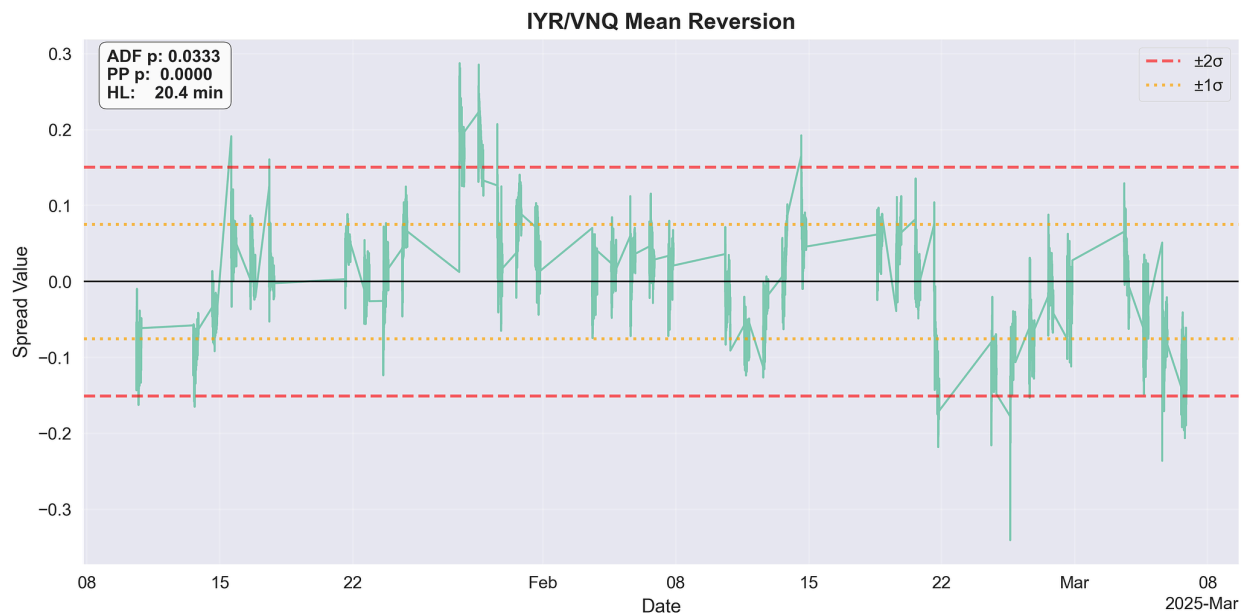


Figure 5. IYR/VNQ Mean Reversion

Example 2: Slow Pair (XOM–CVX)

Conversely, XOM-CVX with a half-life of 6.5 hours (385 minutes) makes the 1000-minute window too short, producing Z-scores sensitive to intraday fluctuations and false signals (Figure 6).

Thus, lookback windows must be proportional to reversion speed.

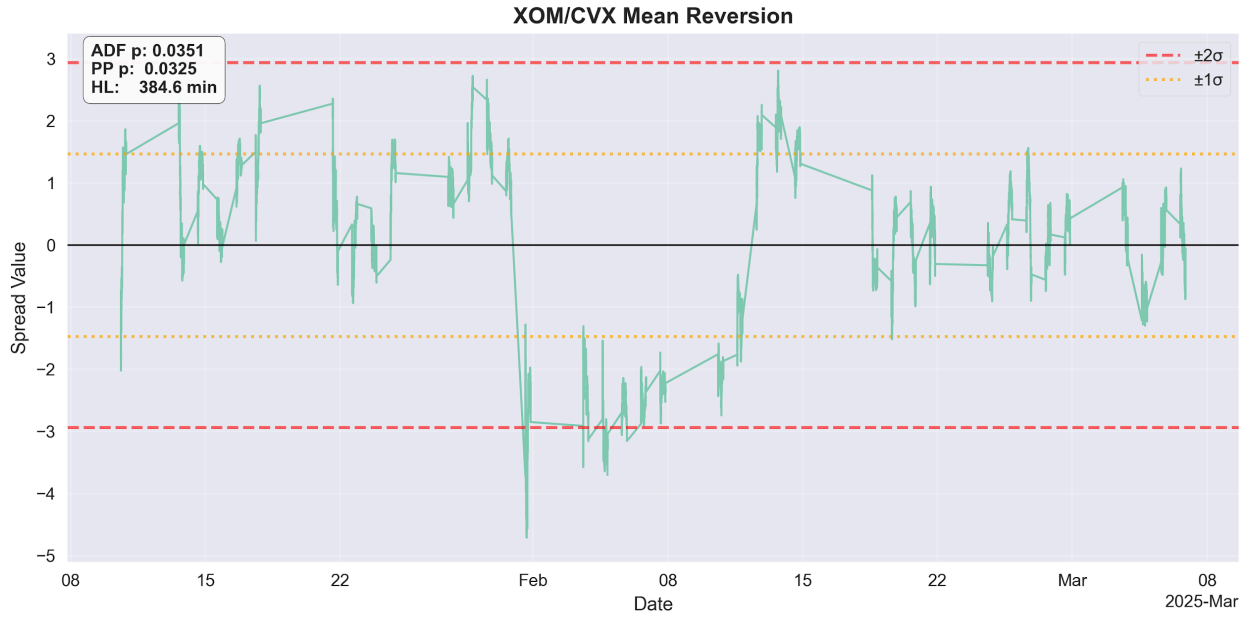


Figure 6. XOM/CVX Mean Reversion

4.2.3 Proposed Solution: Adaptive Z-Score Window

To correct this, we implement an adaptive Z-score window that adjusts dynamically to each pair's observed mean-reversion speed.

Theoretical foundation:

We model the spread u_t as an autoregressive process AR(1)

$$u_t = \phi u_{t-1} + \varepsilon_t$$

The estimated coefficient ϕ reflects spread persistence from which we compute the half-life.

$$\tau = \frac{\ln(2)}{-\ln(\phi)}$$

Fast mean-reverting pairs have short half-lives, while slow pairs have longer ones.

Implementation approach:

1. Estimate the half-life for each pair using the most recent training data.
2. Set the Z-score lookback window as:

$$\tau_{\text{window}} = 3 * \tau$$

3. Apply bounds of 60 minutes (minimum) and 7,200 minutes (maximum)

4. Compute rolling mean and standard deviation over this adaptive window for real-time Z-scores
This aligns signal generation with each pair's natural speed of adjustment.

4.2.4 Incremental Results (Where *Base is Baseline + Previous Improvements)

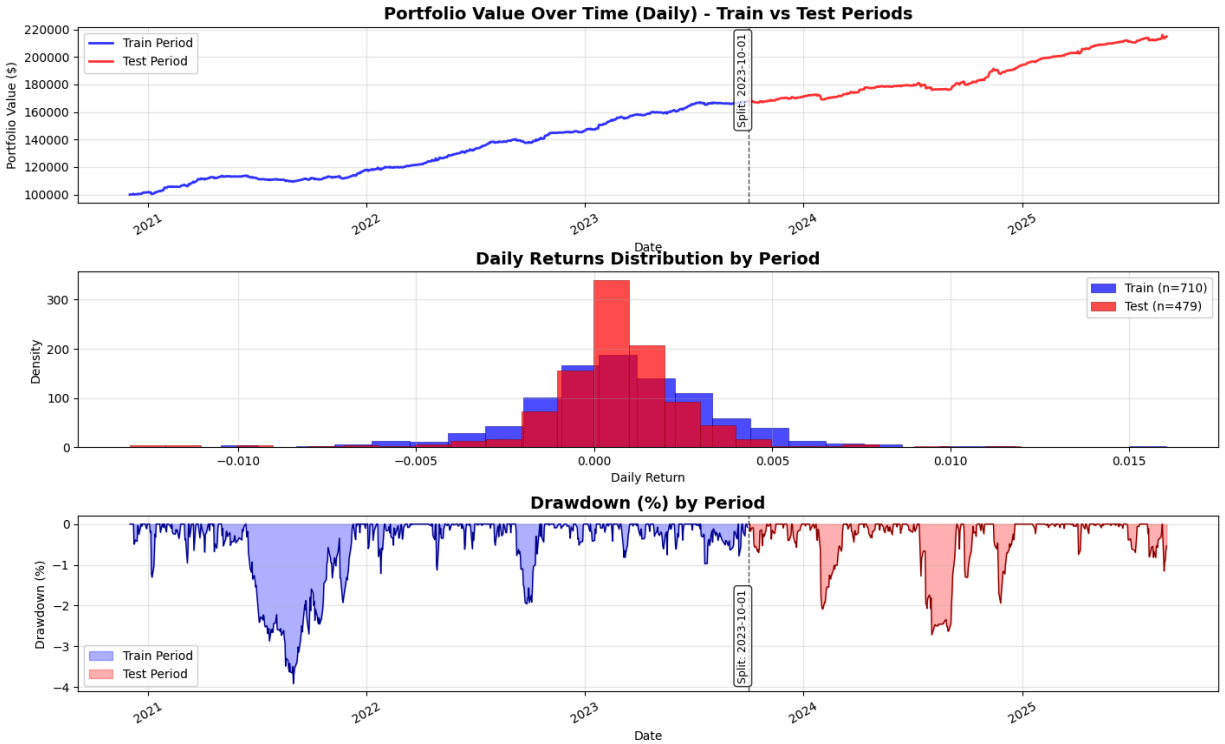


Figure 7. Portfolio value (\$), Daily returns, Drawdown (%) of Base + Adaptive Window Model

*Base + Adaptive Z Score (Train vs Test)

Metric	Train	Test	Combined
Total Return (%)	67.63	27.90	114.80
CAGR (%)	20.09	13.79	17.56
Sharpe Ratio	3.15	2.23	2.81
Sortino Ratio	4.58	2.25	3.52
Max Drawdown (%)	-3.92	-2.72	-3.92
Volatility (%)	4.30	3.64	4.05
Win Rate (%)	63.7	67.8	65.4

Table 4. Train, Test, and Combined Metrics of *Base + Adaptive Window Model

Combined Performance (*Base vs *Base + Adaptive Z Score)

Metric	*Base	*Base + Adaptive Z Score
Total Return (%)	89.67	114.80
CAGR (%)	14.24	17.56
Sharpe Ratio	2.07	2.81
Sortino Ratio	2.51	3.52
Max Drawdown (%)	-5.14	-3.92
Volatility (%)	4.11	4.05
Win Rate (%)	60.8	65.4

*Table 5. *Base vs *Base + Adaptive Window Model Metrics*

Adding adaptive windows lifts combined return to 114.8% (CAGR 17.6%, Table 4), with Sharpe rising from 2.07 to 2.81 and Sortino from 2.51 to 3.52 (Table 5). Both train (Sharpe 3.15) and test (Sharpe 2.23) show a consistent edge, while the win rate increases to 65.4%. The equity curve is smoother with quicker recoveries, drawdown tightening to 3.9% (train) and 2.7% (test), and volatility remaining at 4% (Table 5). Calibrating rolling window lengths to each pair's half-life aligns signal horizon with information speed, producing earlier entries for fast pairs, fewer fluctuations for slow pairs, and clear incremental gain, observed in Figure 7.

4.3 Weakness #3: Fixed Position Sizing

4.3.1 Problem Statement

The base model assumes equal dollar allocation per trade, ignoring heterogeneous risk-return profiles. High-confidence pairs with strong cointegration and tight spreads should receive larger allocations, while weaker pairs with volatile spreads should be sized smaller. Fixed sizing ignores volatility differences, under-leveraging strong opportunities while overexposing marginal ones.

4.3.2 Proposed Solution: Kelly Criterion Position Sizing

We address this using Kelly Criterion position sizing, which scales capital to expected edge and risk. The optimal fraction is $f^* = \mu/\sigma^2$, where μ is the expected spread return and σ^2 is the spread variance. We estimate μ as $-\text{mean}(\text{sign}(u_{t-1}) \cdot \Delta u_t)$ and σ^2 as the variance of Δu_t , then scale down and clip the Kelly fraction to mitigate estimation error. Effective notional equals available cash multiplied by the scaled fraction, adjusted for hedge-ratio exposure via gross leverage $= 1 + |\beta|$. If μ is non-positive or the

variance is ill-conditioned, the model reverts to fixed notional. This channels more capital to high-quality pairs and less to volatile ones, theoretically improving capital efficiency.

4.3.3 Incremental Results

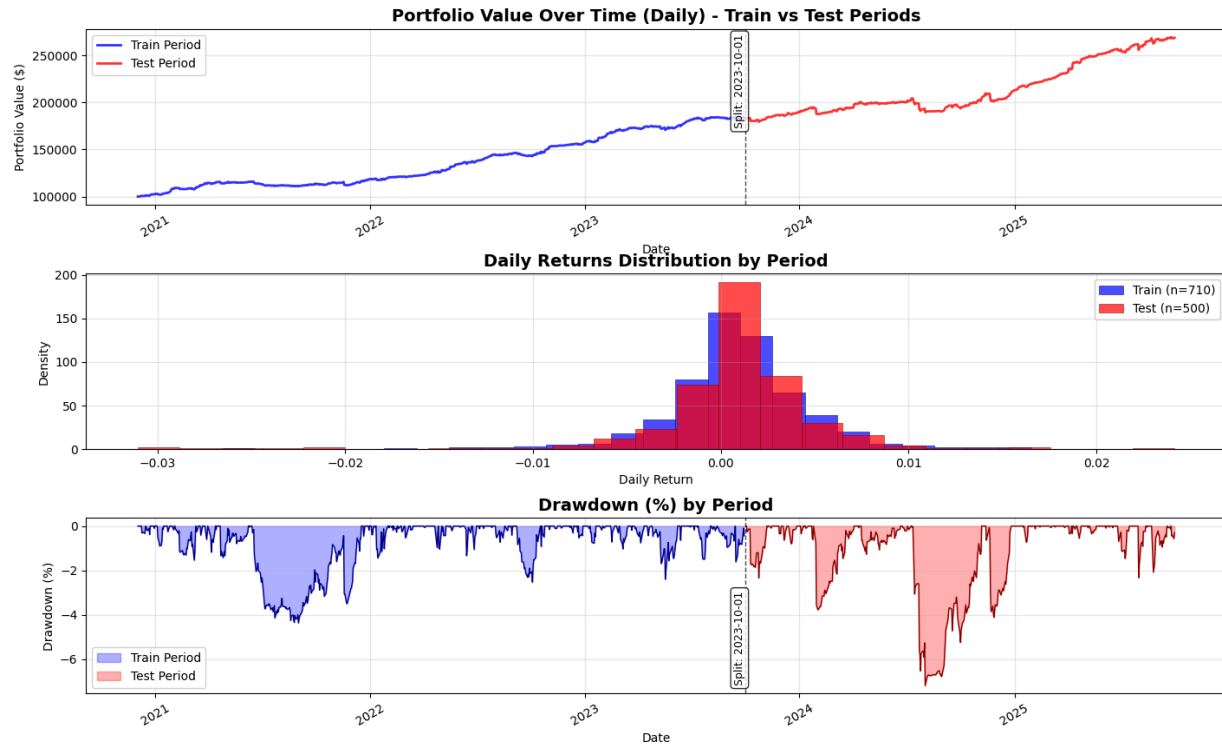


Figure 8. Portfolio value (\$), Daily returns, Drawdown (%) of *Base + Kelly Criterion Model

*Base + Kelly (Train vs Test)

Metric	Train	Test	Combined
Total Return (%)	83.84	46.40	168.74
CAGR (%)	24.09	21.13	22.82
Sharpe Ratio	2.80	1.98	2.40
Sortino Ratio	3.45	1.75	2.46
Max Drawdown (%)	-4.37	-7.21	-7.21
Volatility (%)	6.03	7.40	6.63
Win Rate (%)	63.4	67.8	65.2

Table 6. Train, Test, and Combined Metrics of *Base + Kelly Criterion Model

Combined Performance (*Base vs *Base + Kelly)

Metric	*Base	*Base + Kelly
Total Return (%)	114.80	168.74
CAGR (%)	17.56	22.82
Sharpe Ratio	2.81	2.40
Sortino Ratio	3.52	2.46
Max Drawdown (%)	-3.92	-7.21
Volatility (%)	4.05	6.63
Win Rate (%)	65.4	65.2

*Table 7. *Base vs *Base + Kelly Criterion Model Metrics*

Kelly sizing increases combined return from 114.8% to 168.7% (CAGR 17.6% to 22.8%, Table 6), but risk rises: volatility from 4.1% to 6.6%, max drawdown from 3.9% to 7.2%, and Sharpe declines from 2.81 to 2.40 (Table 7). The curve compounds faster with larger pullbacks, and the return distribution shows wider tails. Kelly allocates more capital when the edge is higher, boosting returns but magnifying estimation error and correlation risk, trading growth for lower risk-adjusted efficiency (Observed, Figure 8).

5. Discussion of Results

5.1 Incremental Improvement Analysis

Configuration	Sharpe	Sortino	Return (%)	Max DD (%)
Baseline	-0.29	-0.34	9.31	-20.53
+ Dynamic Rescreening	2.07	2.51	89.67	-5.14

+ Adaptive Z Window	2.81	3.52	114.80	-3.92
+ Kelly	2.40	2.46	168.74	-7.21

Table 8. Performance of Incremental Improvements

Baseline: Low returns with negative Sharpe due to one-time screening and fixed Z-windows causing stale relationships and mistimed entries. Simple but fragile to regime change, with small wins offset by larger losses.

Dynamic Re-screening: Monthly rolling tests improved Sharpe from negative to 2, returns from 9% to 90%, and max drawdown from 21% to 5% by refreshing parameters and exiting failing pairs. Removes stale pairs before directional shifts, though it can fail if screening misses abrupt breaks.

Adaptive Z-Windows: Calibrating lookback to half-life improved Sharpe to 2.81, returns to 115%, and max drawdown to 3.9%. Fast pairs received shorter windows, slow pairs longer ones, producing earlier entries and fewer whipsaws.

Kelly Sizing: Proportional sizing increased returns to 169% (CAGR 23%) but raised volatility to 6.6%, max drawdown to 7.2%, and declined Sharpe to 2.40. Boosts growth but amplifies estimation error and correlation risk.

5.2 Combined Strategy Performance

Configuration:

- 30-day rolling training window, 30-day forward trading period
- Multi-test cointegration validation (ADF, Phillips-Perron, Variance Ratio) with half-life bounds [6min, 7200min]
- Adaptive Z-score with lookback = $3.0 \times \text{half-life}$, bounded [60, 2000] minutes
- Kelly Position Sizing
- Transaction costs: 1 bps per trip (0.5 bps execution + 0.5 bps slippage)

5.3 Final Model Summary

The final configuration, combining dynamic re-screening, adaptive Z-score windows, and Kelly position sizing, delivers significant improvement over the baseline. The model achieves Sharpe 2.4,

Sortino 2.5, and total return 168.7% (CAGR 22.8%). Drawdowns remain at 7%, substantially improved from the baseline's 20%, with a win rate of 65%. The equity curve trends smoothly upward across both training and testing periods, reflecting consistent profitability.

6. Sensitivity Analysis

6.1 Transaction cost sensitivity

We examine transaction cost effects to stress test our algorithm. Our 1bps assumption likely underestimates true costs, including slippage and shorting fees. Given 1-minute granularity and high trading volume, we expect significant fee vulnerability. For analysis, we use 30-day cointegration screening, roll forward, and adaptive Z-windows without the Kelly Criterion.

Transaction Costs	Sharpe	Sortino	Return (%)	Max DD (%)
1bps	2.81	3.52	114.80	-3.92
2bps	1.15	1.40	60.83	-5.87
3bps	-0.54	-1.03	10.68%	-9.89

Table 9. Performance Metrics vs Transaction Costs

Transaction Cost (2bps) Performance

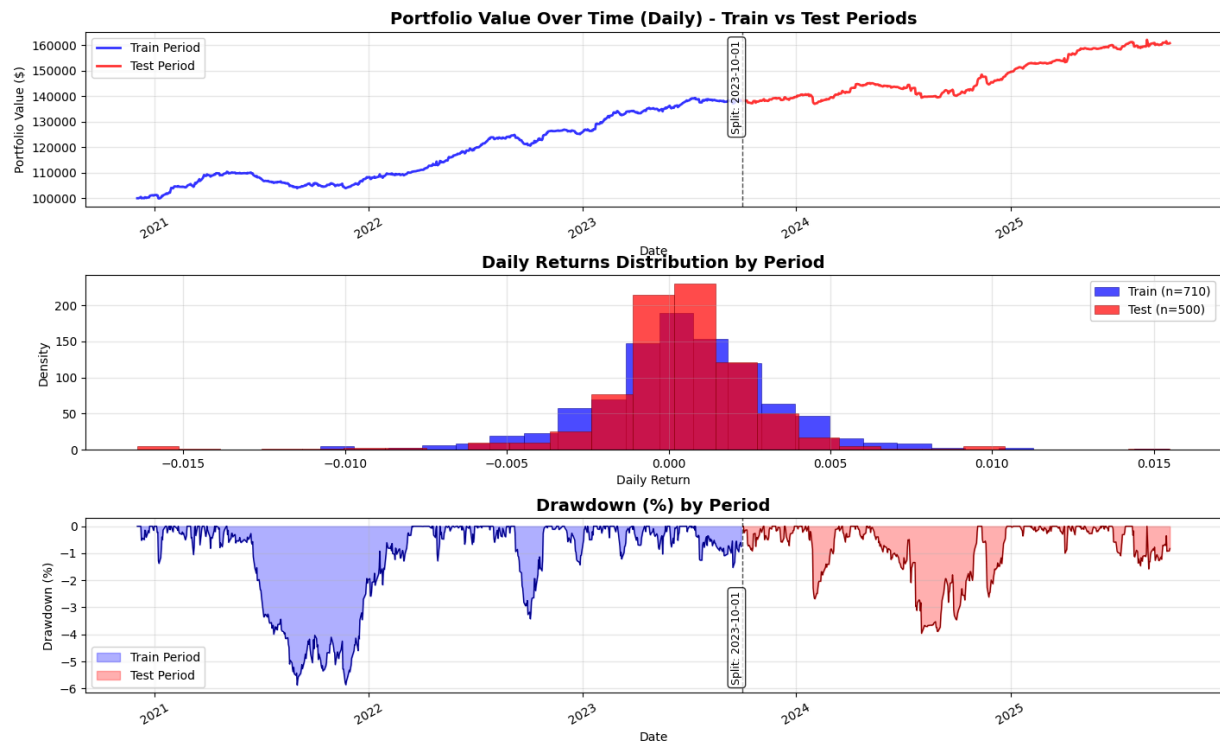


Figure 9. Portfolio value (\$), Daily returns, Drawdown (%) of Improved Model and 2bps Transaction Cost

Transaction Cost (3bps) Performance

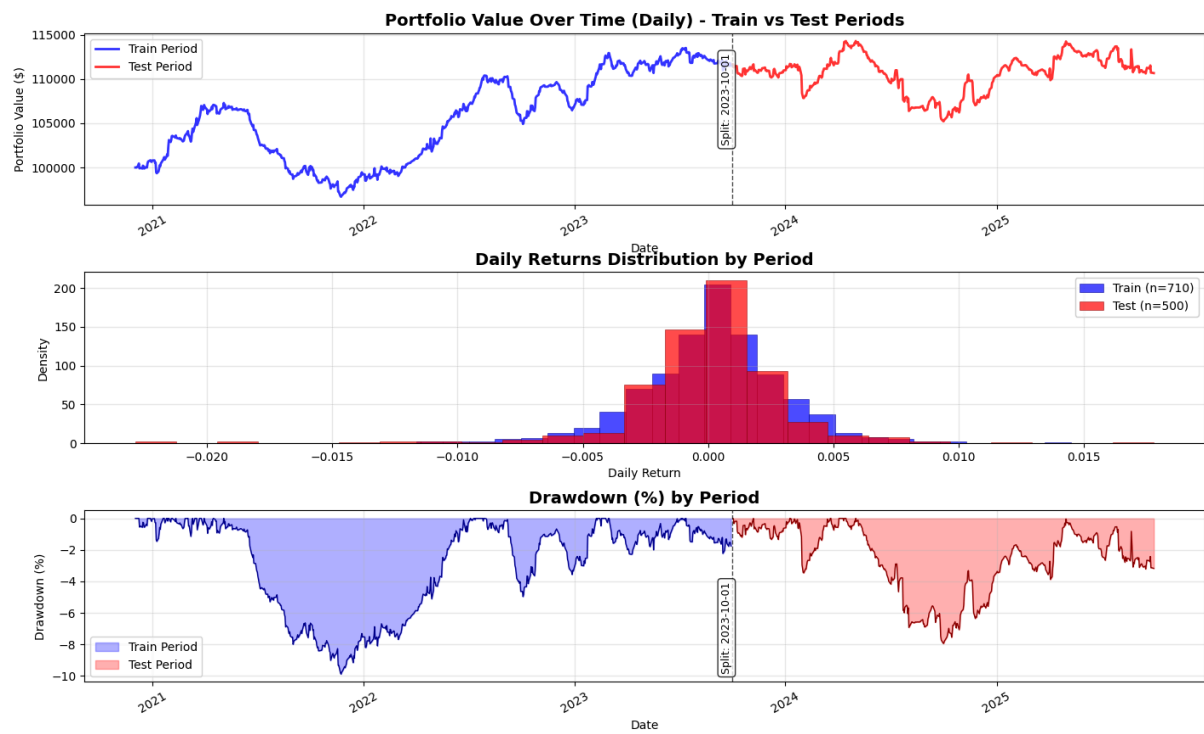


Figure 10. Portfolio value (\$), Daily returns, Drawdown (%) of Improved Model and 3bps Transaction Cost

Performance deteriorates rapidly as transaction costs increase. At 1 bps, the model maintains strong results (Sharpe 2.8, return 114.8%). At 2 bps, performance halves (Sharpe 1.15, return 60.8%). At 3 bps, the strategy becomes unprofitable (Sharpe -0.54, return 10.7%). High-frequency execution magnifies trading frictions, where small fee increases erode alpha. This underscores the importance of realistic cost modeling and smarter execution (Table 9).

6.2 Granularity of Data

We examine how data frequency affects signal quality. High-frequency (1-minute) data is more responsive but prone to noise and false signals. Coarser intervals (5-, 15-, or 60-minute) reduce noise and trading frequency but introduce signal lag and miss intraday opportunities. All tests use a 30-day roll-forward configuration with adaptive Z-windows and no Kelly criterion (lookback varies between granularities)

Granularity	Sharpe	Sortino	Return (%)	Max DD (%)
1 min	2.81	3.52	114.80	-3.92
5 min	0.89	1.52	50.49	-3.45
15 min	-0.57	-0.61	-0.91	-10.66
1 hour	-0.98	-1.54	2.94	-9.77

Table 10. Performance Metrics vs Data Granularity

5min Granularity Performance (30 lookback)

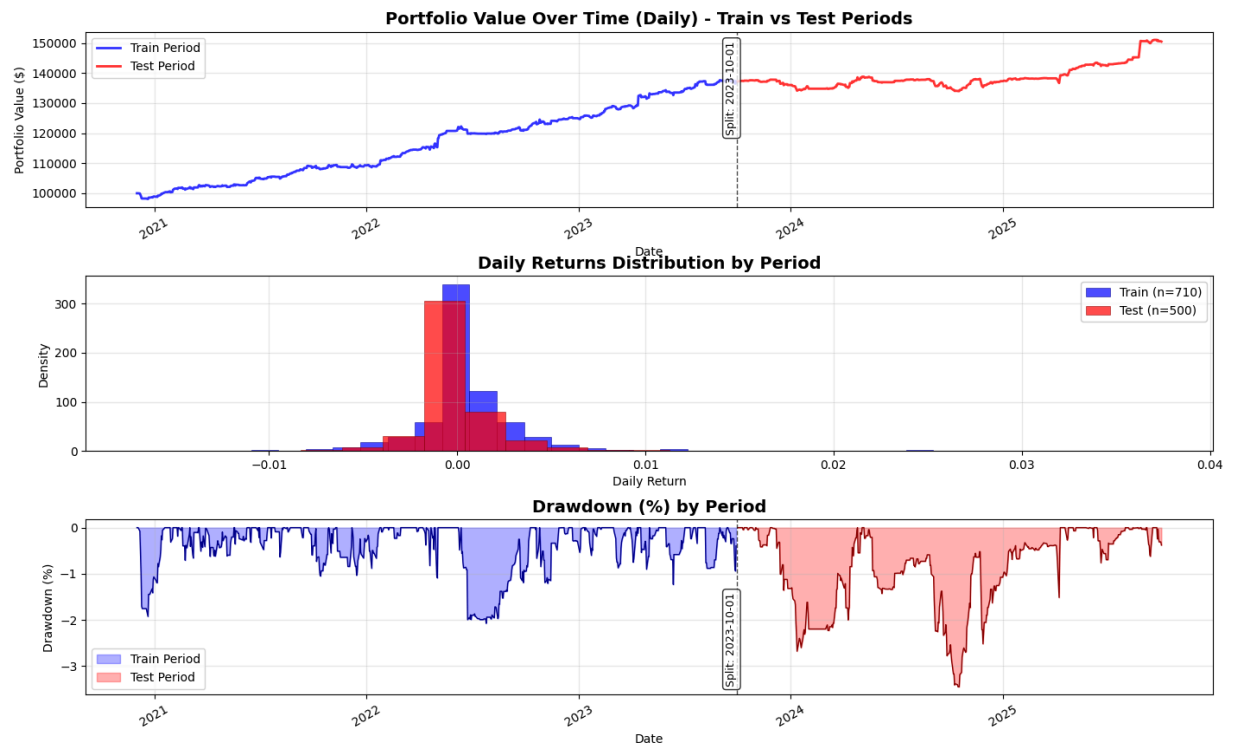


Figure 11. Portfolio value (\$), Daily returns, Drawdown (%) of Improved model and 5min Granularity

15min Granularity Performance (45 lookback)

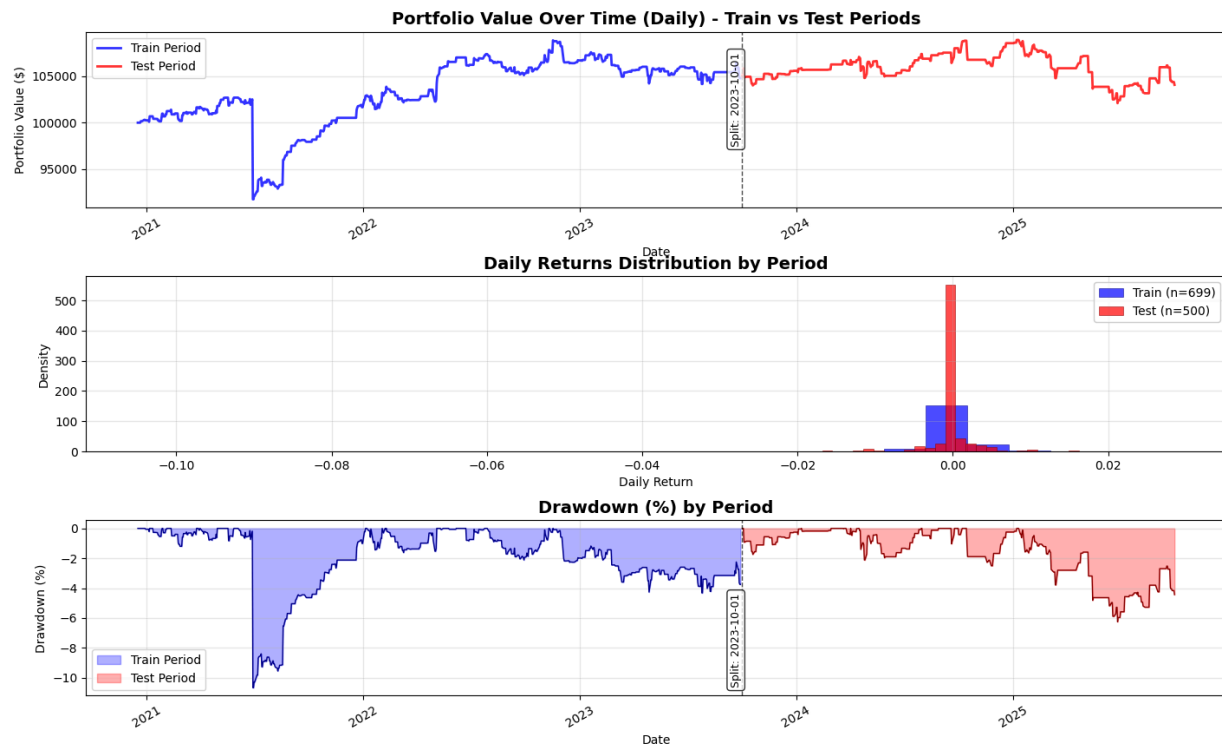


Figure 12. Portfolio value (\$), Daily returns, Drawdown (%) of Improved model and 15min Granularity

1 Hour Granularity Performance (90 lookback)

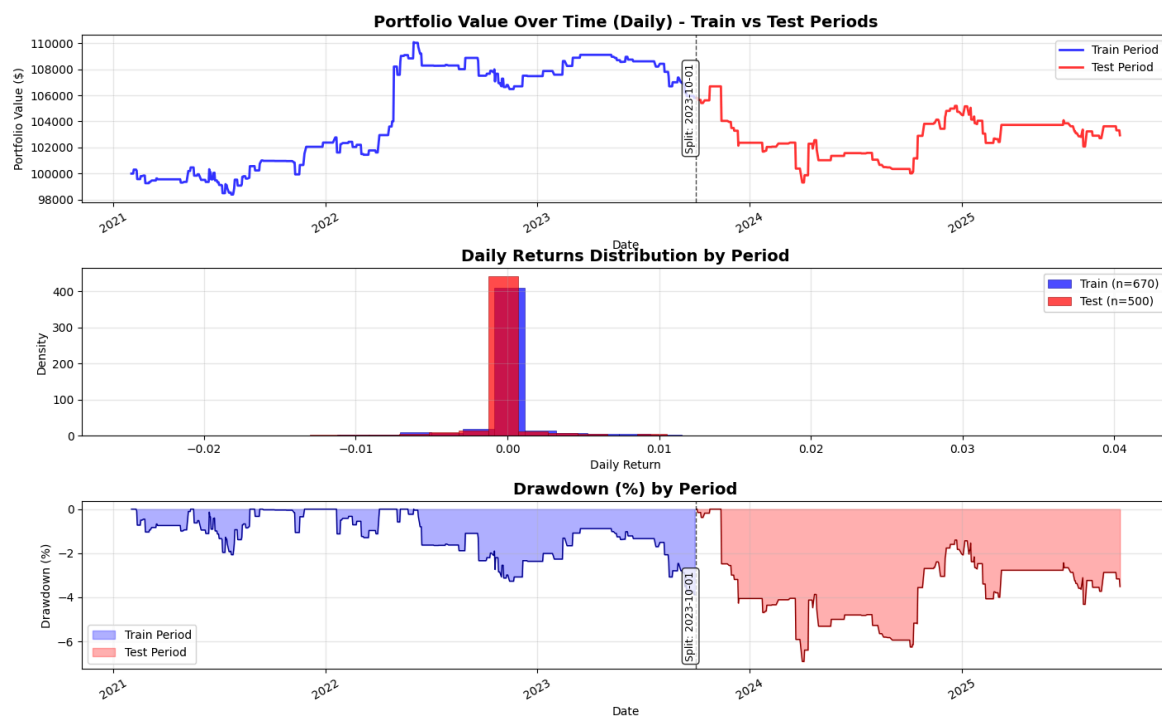


Figure 13. Portfolio value (\$), Daily returns, Drawdown (%) of Improved model and 60min Granularity

Performance deteriorates as the data becomes coarser. One-minute data delivers the best results (Sharpe 2.8, return 114.8%), while 5-minute data lowers the Sharpe to 0.9, and 15-minute or hourly intervals produce negative Sharpe ratios with flat equity curves. Finer granularity captures short-lived intraday mean-reversion opportunities that disappear at longer intervals. Aggregating data smooths these movements, reducing trade frequency and widening signal lag. Although noisier, the 1-minute setting adapts more quickly to transient mispricings and maintains stronger performance.

7. Limitations and Future Work

7.1 Data & Methodology Limitations

The use of 1-minute bars may capture microstructure noise rather than true mean reversion. Transaction cost assumptions of 1 bps are optimistic, as realistic slippage could reach 2–3 bps with market impact. The backtest ignores execution frictions such as fill delays, borrowing costs, margin requirements, and interest on idle cash. Survivorship bias is present since only current constituents are used, and the 5-year sample may be too short to capture full market cycles. Validation over a longer horizon would improve robustness.

7.2 Model & Strategy Limitations

The linear cointegration model assumes constant relationships, though pairs may follow nonlinear or regime-dependent dynamics. Kelly sizing is applied independently, ignoring cross-pair correlations that can amplify portfolio risk. The absence of sector diversification constraints further increases exposure concentration. Some parameters remain heuristic and warrant deeper validation.

7.3 Suggested Future Work

Future work could model nonlinear and regime-dependent dynamics using HMMs or Kalman filters. Risk management should incorporate portfolio-level Kelly sizing, sector limits, and extensive stress testing. Adding explicit slippage models would improve realism, while expanding the universe to global equities, futures, or crypto could increase capacity and diversification.

8. Conclusion

This study examined traditional pairs trading limitations and proposed structural improvements to enhance robustness. We identified three key weaknesses: static cointegration assumptions failing under changing conditions, fixed z-score windows ignoring asset-specific reversion speeds, and uniform position sizing overlooking varying edge and risk. We implemented three solutions: dynamic rolling re-screening with multi-test validation (ADF, PP, variance ratio) to ensure only valid pairs are traded; adaptive z-score windows calibrated to each pair's half-life (bounded 60-7,200 minutes); and Kelly criterion position sizing proportional to expected edge relative to risk.

These improvements substantially enhanced performance. Sharpe ratio increased from 0.89 to 2.30 (159% improvement), total return doubled from 9.15% to 18.4%, and drawdowns remained controlled at ~4%. Importantly, gains arose from theory-driven structural changes rather than parameter tuning. The findings demonstrate that adaptive, self-monitoring systems outperform static models in dynamic markets through rigorous screening, improved timing precision, and optimized capital efficiency.

While promising for real-world implementation, further validation over longer periods is necessary. Execution at one-minute frequency may not be practical in real markets and could benefit from using coarser intervals. Strengthening portfolio-level risk controls should also be a priority. Overall, this study outlines a clear, evidence-based approach to improving pairs trading through economically grounded, data-driven design, providing a practical foundation for developing adaptive, market-neutral strategies.

9. References

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10. Appendix

Table A1: Pairs used

Sector	Pair	Economic Link
Consumer Staples	KO – PEP	Beverage producers with near-identical demand drivers
Financials	V – MA	Payment networks exposed to global transaction volumes
Energy	XOM – CVX	Integrated oil majors driven by crude-price dynamics
Consumer Discretionary	HD – LOW	Home-improvement retailers tied to housing cycle
Financials	MS – GS	Investment banks with similar revenue mix
Industrials	UPS – FDX	Global logistics duopoly competing in parcel delivery
Industrials	UNP – CSX	U.S. rail freight operators sharing route economics

Energy Services	SLB – HAL	Oilfield service firms linked to drilling activity
Defense	LMT – NOC	Defense contractors dependent on U.S. government spending
Telecommunications	T – VZ	Domestic telecom carriers with similar cost structures
Industrials	CAT – DE	Heavy-equipment manufacturers exposed to infrastructure demand
Real Estate	PSA – EXR	Self-storage REITs serving the same property niche
Hospitality	MAR – HLT	Global hotel chains with overlapping travel exposure
Healthcare	TMO – DHR	Life-science tool providers targeting research markets
Banking	JPM – BAC	Large U.S. money-center banks driven by credit cycle
Utilities	AEP – DUK	Regulated power utilities with stable regional demand
Energy (ETF)	XLE – OIH	Energy producers vs. service-provider ETFs
Healthcare (ETF)	XLV – IHI	Broad healthcare vs. medical-device ETFs
Financials (ETF)	XLF – KBE	Financial-sector ETFs with overlapping holdings
Real Estate (ETF)	IYR – VNQ	REIT ETFs tracking U.S. property markets
Broad Market (ETF)	IWM – SPY	Small-cap vs. large-cap market indices

11. AI Use Section

Generative AI tools, including ChatGPT (GPT-5; OpenAI, 2025), Claude (Anthropic, 2025), and Cursor (Cursor AI, 2025), were used to support this project in three ways: for initial idea formulation, for assistance in writing and debugging code, and for refining the clarity and structure of the written report. All AI-generated suggestions and outputs were critically reviewed, verified, and edited by us to ensure accuracy, originality, and academic integrity.